

CHAPTER 14

RANK METHODS

14.1 INTRODUCTION

This chapter is devoted to statistical tests applicable to data measured on ordinal scales only and more precisely, when only ordinal relations are taken into account. These methods³² fall under the general heading “nonparametric statistics,” and they involve techniques based on ranks, which offer several advantages. While the data sampled must be from a continuous distribution, the type of the distribution is irrelevant. Nonparametric methods are almost as powerful as normal theory methods. They can therefore be applied when a population distribution is not normal (e.g., it is skewed) or if the sample size is not large enough for testing normality.

■ EXAMPLE 14.1

Suppose that four observations randomly selected from some population are $x_1 = 8, x_2 = 7, x_3 = 3, x_4 = 15$, and five observations from another population are $y_1 = 5, y_2 = 4, y_3 = 0, y_4 = 6, y_5 = 1$. If the data are expressed on the ordinal (but not any higher) scale, these data contain the same information as such data as $x_1 = 1,000, x_2 = 32, x_3 = 30, x_4 = 1,001, y_1 = 35, y_2 = 31, y_3 = -5, y_4 = 36, y_5 = 0$. In effect all relevant information is contained in the sequence

YYXYYYXXX,

which tells us that the observations from the first sample occupy places 3, 7, 8, and 9 in the ordered combined sample.

³²See Hollander and Wolfe (1999) for an exhaustive presentation.

Methods based on ranks *should* be used for the data measured on an interval scale. They, however, *can* be used for the data measured on an interval or ratio scale. In other words, one can always use methods for a weaker scale.

The assumption of continuity is what distinguishes the methods of this chapter from the methods of categorical data that sometimes are also of an ordinal character (discussed in Chapter 15) and exemplified by questionnaire responses such as “strongly agree,” “agree,” “neutral,” “disagree,” “strongly disagree.” The essential difference is that under the assumption of a population distribution being continuous, we can disregard ties in our theoretical considerations. Practically, it means that ties in the data are rather exceptional.

We begin with the study of the behavior of empirical cumulative distribution functions (cdf's). The empirical cdf converges almost surely and uniformly to the cdf of the underlying random variable. This is a general fact, and it is true regardless of whether or not the random variable is continuous. If we restrict the analysis to the continuous case, we can construct a test for the hypothesis that the random sample $X_1, \dots, X_n$ comes from a population with a specific continuous cdf.

Next we will consider the two-sample problem, tests of the hypothesis that two random samples $(X_1, \dots, X_m)$ and $(Y_1, \dots, Y_n)$ come from the same continuous distribution. We will also introduce tests for randomness and present procedures for testing hypotheses about population medians in one, two, and more than two populations. Last, we will present an analysis of variance (ANOVA) type of inference in case of response distribution that is other than normal.

14.2 GLIVENKO–CANTELLI THEOREM

Let $\mathbf{X}_n = (X_1, \dots, X_n)$ be a random sample from some population. Specifically, this means that $X_1, \dots, X_n$ is the initial fragment of a sequence $\mathbf{X}_\infty = \{X_i, i = 1, 2, \dots\}$ of iid random variables. We let F denote the common cdf of X_i 's so that $F(t) = P\{X_i \leq t\}$. At the moment, we can make no assumptions about the nature of F . We only know that F is an arbitrary nondecreasing right-continuous function, satisfying

$$\lim_{t \rightarrow \infty} F(t) = 1, \quad \lim_{t \rightarrow -\infty} F(t) = 0$$

(see Chapter 5). For any n and t , we define the empirical cdf of the sample $X_1, \dots, X_n$ as

$$F_n(t) = \frac{\text{number of data values that do not exceed } t}{n}.$$

Thus $F_n(t)$ is a step function that increases by $1/n$ (or by a multiple of $1/n$) at each point of the sample. It is important to keep in mind that $F_n(t)$ is a random function. It depends formally on the sequence $\mathbf{X}_\infty$ but in fact only on the first n observations.

If we fix the value t , then $nF_n(t)$ is a random variable, equal to the number of X_i 's among the first n observations, satisfying the condition $X_i \leq t$. Thus $nF_n(t)$ has a binomial distribution with parameters n and $p = P\{X_i \leq t\} = F(t)$. Consequently, by Theorem 9.5.20, for every t ,

$$F_n(t) = \frac{nF_n(t)}{n} \rightarrow F(t) \quad (14.1)$$

with probability 1.

To appreciate the meaning of this result, and the meaning of its extension below, let us write explicitly $F_n(t, \mathbf{X})$, where $\mathbf{X} = \mathbf{X}_\infty$. Since $\mathbf{X}$ is the element of the sample space corresponding to sampling an infinite sequence of values of iid random variables, each with cdf F , the phrase “with probability 1” or “almost certainly” means “for all sequences $\mathbf{X}$, except sequences in a set $\mathcal{N}$ with $P(\mathcal{N}) = 0$.”

Specifically, (14.1) means that for every t there exists a set $\mathcal{N}_t$ of sequences $\mathbf{X}$ such that $P(\mathcal{N}_t) = 0$ and if $\mathbf{X} \notin \mathcal{N}_t$, then

$$\lim_{n \rightarrow \infty} F_n(t, \mathbf{X}) = F(t). \quad (14.2)$$

The following theorem is an extension of this result to the convergence for all t at once, and uniform:

Theorem 14.2.1 (Glivenko–Cantelli) *As $n \rightarrow \infty$, we have*

$$\sup_t |F_n(t) - F(t)| \rightarrow 0$$

with probability 1.

Proof: The theorem means that one can find a set $\mathcal{N}$ of sequences $\mathbf{X}$ such that $P(\mathcal{N}) = 0$ and $\sup_t |F_n(t, \mathbf{X}) - F(t)| \rightarrow 0$ as $n \rightarrow \infty$ for all $\mathbf{X} \notin \mathcal{N}$. We show first that such a common set $\mathcal{N}$ exists for convergence $F_n(t)$ to $F(t)$ for all t (convergence being not necessarily uniform). Let A be the set of all points at which F is discontinuous. Since F is nondecreasing and continuous on the right, the condition $t \in A$ means that $F(t) - F(t-) > 0$, where

$$F(t-) = \lim_{\tau \rightarrow t-} F(\tau) = \sup_{\tau < t} F(\tau).$$

If $t \in A$, we define the random variables

$$U_n(t) = U_n(t, \mathbf{X}) = \begin{cases} 1 & \text{if } X_n = t \\ 0 & \text{if } X_n \neq t, \end{cases} \quad (14.3)$$

so that $E[U_n(t)] = P\{U_n(t) = 1\} = F(t) - F(t-)$. Clearly, the random variables $U_1(t)$, $U_2(t)$, $\dots$ are iid, and by Theorem 9.5.20, we have

$$\frac{1}{n} \sum_{j=1}^n U_j(t, \mathbf{X}) \rightarrow F(t) - F(t-)$$

for all $\mathbf{X} \notin \mathcal{N}_t^*$, where $P(\mathcal{N}_t^*) = 0$. Here, $\mathcal{N}_t^*$ is the “exceptional” set chosen for specific t .

Clearly, the set A is at most countable. Indeed, if $A_k \subset A$ is the set of points $t \in A$ with $F(t) - F(t-) \geq 1/k$, then A_k has at most k elements because of monotonicity of F and the condition $F(\infty) - F(-\infty) = 1$. Since $A = \bigcup_k A_k$, the set A is at most countable, and the condition (14.3) holds for all $\mathbf{X} \notin \bigcup_{t \in A} \mathcal{N}_t^*$, with

$$P\left(\bigcup_{t \in A} \mathcal{N}_t^*\right) \leq \sum_{t \in A} P(\mathcal{N}_t^*) = 0.$$

Now let Q be the set of all rational t (or any other countable dense subset of real line). We may assert that $F_n(t) \rightarrow F(t)$ for all $t \in Q$ with probability 1, since by (14.2) we have $F_n(t, \mathbf{X}) \rightarrow F(t)$ for all t if $\mathbf{X} \notin \bigcup_{t \in Q} \mathcal{N}_t$, and again $P(\bigcup_{t \in Q} \mathcal{N}_t) \leq \sum_{t \in Q} P(\mathcal{N}_t) = 0$.

Thus we showed that with probability 1 we have $F_n(t) \rightarrow F(t)$ for all $t \in Q$ and $F_n(t) - F_n(t-) \rightarrow F(t) - F(t-)$ for all $t \in A$. We will next show that with probability 1 we also have $F_n(t) \rightarrow F(t)$ for all $t \notin A$, and that the convergence is uniform. These last statements do not involve any probability considerations: They are true for any $\mathbf{X} \notin \left(\bigcup_{t \in A} \mathcal{N}_t^*\right) \cup \left(\bigcup_{t \in Q} \mathcal{N}_t\right)$, so one may suppress the dependence on $\mathbf{X}$ and consider a fixed sequence of cdf's $F_n(\cdot)$.

Thus let t be a continuity point of F , and let $t \notin Q$. We want to show that $F_n(t) \rightarrow F(t)$. Let $\epsilon > 0$, and let $t_1, t_2 \in Q$ satisfy the relations $t_1 < t < t_2$ and

$$F(t_2) - F(t_1) < \frac{\epsilon}{2} \quad (14.4)$$

(which is possible because t is a continuity point of F). Next we choose N such that for $n \geq N$,

$$|F_n(t_1) - F(t_1)| < \frac{\epsilon}{2} \quad \text{and} \quad |F_n(t_2) - F(t_2)| < \frac{\epsilon}{2}. \quad (14.5)$$

This is possible because $F_n(t_i) \rightarrow F(t_i)$ as $n \rightarrow \infty$ for $i = 1, 2$. By monotonicity of F we have $F(t_1) \leq F(t) \leq F(t_2)$; hence by (14.4),

$$F(t_2) - \frac{\epsilon}{2} < F(t) < F(t_1) + \frac{\epsilon}{2}.$$

Using (14.4), (14.5), and monotonicity of $F_n(\cdot)$, we can write for $n \geq N$,

$$F(t) \leq F(t_2) \leq F(t_1) + \frac{\epsilon}{2} \leq \left(F_n(t_1) + \frac{\epsilon}{2}\right) + \frac{\epsilon}{2} \leq F_n(t) + \epsilon$$

and also

$$F(t) \geq F(t_2) - \frac{\epsilon}{2} \geq \left(F_n(t_2) - \frac{\epsilon}{2}\right) - \frac{\epsilon}{2} \geq F_n(t) - \epsilon,$$

which gives $|F_n(t) - F(t)| \leq \epsilon$. It remains to prove that the convergence $F_n(t) \rightarrow F(t)$ is uniform, that is, $\sup_t |F_n(t) - F(t)| \rightarrow 0$ as $n \rightarrow \infty$. The proof is after Chung (2001).

Now assume the contrary. There exists $\epsilon_0 > 0$, and sequences n_k and t_k such that $|F_{n_k}(t_k) - F(t_k)| > \epsilon_0$ for all t . Since $F_n(\cdot)$ and $F(\cdot)$ are cdf's, we cannot have $t_k \rightarrow +\infty$ or $t_k \rightarrow -\infty$; hence the sequence $\{t_k\}$ is bounded. Without loss of generality we can assume that $t_k \rightarrow t^*$.

Moreover, there exists either a subsequence of $\{t_k\}$ which converges to t^* monotonically from below or monotonically from above. Similarly, since $|F_{n_k}(t_k) - F(t_k)| > \epsilon_0$, there exists either a subsequence at which $F_{n_k}(t_k) > F(t_k) + \epsilon_0$ or a subsequence at which $F_{n_k}(t_k) < F(t_k) - \epsilon_0$. Restricting the analysis to these subsequences, we distinguish four cases:

1. $t_k \uparrow t^*, t_k < t^*, F_{n_k}(t_k) > F(t_k) + \epsilon_0$.
2. $t_k \uparrow t^*, t_k < t^*, F_{n_k}(t_k) < F(t_k) - \epsilon_0$.
3. $t_k \downarrow t^*, F_{n_k}(t_k) > F(t_k) + \epsilon_0$.
4. $t_k \downarrow t^*, F_{n_k}(t_k) < F(t_k) - \epsilon_0$.

We will select $t', t'' \in Q$ such that $t' < t^* < t''$.

In case 1, using the monotonicity of F_{n_k} and F , we can write, for all k sufficiently large,:

$$\begin{aligned} \epsilon_0 &< F_{n_k}(t_k) - F(t_k) \leq F_{n_k}(t^* -) - F(t') \\ &\leq F_{n_k}(t^* -) - F_{n_k}(t^*) + F_{n_k}(t'') - F(t'') + F(t'') - F(t'). \end{aligned}$$

If we let $k \rightarrow \infty$, the difference $F_{n_k}(t^* -) - F_{n_k}(t^*)$ converges to $-(F(t^*) - F(t^* -))$, and the difference $F_{n_k}(t'') - F(t'')$ converges to 0. Letting $t'' \downarrow t^*, t' \uparrow t^*$ along values in Q , the last difference converges to $F(t^*) - F(t^* -)$. Thus the right-hand side can be made as small as possible, which gives a contradiction.

In case 2, we have

$$\epsilon_0 < F(t_k) - F_{n_k}(t_k) < F(t^* -) - F_{n_k}(t') \leq F(t^* -) - F(t') + F(t') - F_{n_k}(t').$$

Letting $k \rightarrow \infty$, we obtain $F(t') - F_{n_k}(t') \rightarrow 0$, and letting next $t' \uparrow t^*$, we obtain $F(t^* -) - F(t') \rightarrow 0$, which gives another contradiction.

In case 3, we have

$$\begin{aligned} \epsilon_0 &< F_{n_k}(t_k) - F(t_k) < F_{n_k}(t'') - F(t^*) \\ &\leq F_{n_k}(t'') - F_{n_k}(t') + F_{n_k}(t') - F(t') + F(t') - F(t^*). \end{aligned}$$

Now $F_{n_k}(t'') - F_{n_k}(t') \rightarrow F(t'') - F(t')$, while $F_{n_k}(t') - F(t') \rightarrow 0$. We let $t'' \downarrow t^*$ and use the continuity of F on the right, which results in the term $F(t'') - F(t')$ converging to $F(t^*) - F(t')$. So we again have a contradiction.

Finally, in case 4, we can write

$$\begin{aligned}\epsilon_0 &\leq F(t_k) - F_{n_k}(t_k) \leq F(t'') - F_{n_k}(t^*) \\ &\leq F(t'') - F(t') + F(t') - F_{n_k}(t') + F_{n_k}(t^*) - F_{n_k}(t^*).\end{aligned}$$

As $k \rightarrow \infty$ we have $F(t') - F_{n_k}(t') \rightarrow 0$, while $F_{n_k}(t^*) - F_{n_k}(t^*) \rightarrow F(t^*) - F(t^*)$. Letting now $t' \uparrow t^*$ and $t'' \downarrow t^*$, we have $F(t'') - F(t') \rightarrow F(t^*) - F(t^*)$, which also leads to contradiction. $\square$

PROBLEMS

14.2.1 Let $F(t) = 0$ for $t < 0$, $F(t) = p$ for $0 \leq t < 1$, and $F(t) = 1$ otherwise. Use the central limit theorem to evaluate directly the distribution of $\sup_t |F_n(t) - F(t)|$, and show that $F_n(t)$ tends to $F(t)$ almost surely and uniformly in t .

14.2.2 Let X be a random variable such that $P\{X = a\} = 1/2$, $P\{X = b\} = 1/3$ and $P\{X = c\} = 1/6$. Let $X_1, X_2, \dots, X_{200}$ be a random sample from the distribution of X . Suppose that among the first 100 observation of X_i , 55 were equal to a and 38 were equal to b . Among the next 100 observations, 51 are equal to a , and 30 are equal to b . Determine $\sup_t |F_{200}(t) - F(t)|$ and $\sup_t |F_{100}(t) - F(t)|$ in all six cases $a < b < c$, $a < c < b$, $b < a < c$, and so on, as well as in all cases such as $b = c < a$, $a = b < c$, $\dots$, and $a = b = c$.

14.3 KOLMOGOROV-SMIRNOV TESTS

One-Sample Kolmogorov-Smirnov Test

Let

$$D_n = \sup_t |F_n(t) - F(t)|. \quad (14.6)$$

The distance D_n converges a.s. to zero if the F_n 's are empirical cdf's of random samples drawn from distribution F . Research on the rate of this convergence, that is, finding constants $c_n \rightarrow \infty$ such that the sequence of random variables $\{c_n D_n, n \geq 1\}$ has a limiting distribution, led to a remarkable discovery, due to Kolmogorov and Smirnov: if F is continuous, then this limiting distribution exists for $c_n = \sqrt{n}$, and moreover, it does not depend on F .

The following theorem specifies this limiting distribution, and serves as a foundation for a test of the hypothesis that a random sample comes from a specific distribution. We will omit the proof here.

Theorem 14.3.1 (Kolmogorov and Smirnov) *Let $X_1, X_2, \dots$ be iid random variables, with a continuous cdf F . Then for every $z > 0$,*

$$\lim_{n \rightarrow \infty} P\{\sqrt{n} D_n \leq z\} = Q(z),$$

where

$$Q(z) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 z^2}. \quad (14.7)$$

The function $Q(z)$ is a cdf of a continuous distribution called a *Kolmogorov distribution*. The values of this cdf are given in Table B2.

Suppose now that we want to test the hypothesis

$$H_0 : F = F_0 \quad \text{against the alternative} \quad H_1 : F \neq F_0,$$

where the hypothetical cdf F_0 is continuous. Under the null hypothesis, for large n , the distribution of $\sqrt{n}D_n$ is given by $Q(x)$. If the true distribution of X_i 's is F^* , we can write

$$\begin{aligned} \sup_t |F_n(t) - F_0(t)| &\leq \sup_t \{|F_n(t) - F^*(t)| + |F^*(t) - F_0(t)|\} \\ &\leq \sup_t |F_n(t) - F^*(t)| + \sup_t |F^*(t) - F_0(t)| \\ &= D_n + \sup_t |F^*(t) - F_0(t)|. \end{aligned}$$

Upon multiplication by $\sqrt{n}$, the first term, $\sqrt{n}D_n$, has the limiting distribution (14.7), while the second term $\sqrt{n} \sup_t |F^*(t) - F_0(t)|$ tends to infinity if $F^*(t) \neq F_0(t)$.

The last property means that the test should reject H_0 if the observed value of the statistic $\sqrt{n}D_n$ exceeds the critical value determined from the right tail of the distribution (14.7). This test has—in the limit—power 1 against *any* alternative.

■ EXAMPLE 14.2

A small town had 30 fires last year: on January 5 and 18, February 3, 4, 21, and 26, March 5, 10, and 13, April 6, May 16 and 25, June 19, July 10 and 21, August 12 and 15, September 1, 8, and 21, November 2, 6, 7, 19, and 29, December 3, 9, 12, 17, and 24. Are these data consistent, on the 0.05 level, with the hypothesis that the occurrences of fires follow a Poisson process?

SOLUTION. One of the solutions may be based on the fact that if the fires form a Poisson process, then they occur throughout the year according to the uniform distribution. The data are summarized in Table 14.1

Since $F_n(\cdot)$ is a step function, it suffices to inspect only the differences $|F(x_i) - F_n(x_i)|$ and $|F(x_i) - F_n(x_{i-1})|$. The largest difference in Table 14.1 is $F(x_{21}) - F_{30}(x_{20}) = 0.838 - 0.667 = 0.171$, and $\sqrt{n}D_n = \sqrt{30} \times 0.171 = 0.937$. Since (see Table A2) $P\{\sqrt{n}D_n > 0.9\} = 1 - 0.6073 = 0.3927$, there is no reason to reject the null hypothesis that the fires form a Poisson process.

It is important to realize that the Kolmogorov–Smirnov test applies only when the null hypothesis specifies completely the distribution F . The test cannot be used in cases where the null hypothesis specifies only the distribution type but does not provide the values of the parameters. If one estimates parameters from the sample to obtain a specific distribution F^* , $F^*(x)$ is then a *random* cdf that depends on the sample, and consequently the limiting distribution of $\sqrt{n}D_n^* = \sqrt{n} \sup_x |F_n(x) - F^*(x)|$ will *not* be given by formula (14.7).

■ EXAMPLE 14.3

In this application of the Kolmogorov–Smirnov test we will regard points distributed uniformly on a circle. The quality of the estimation depends heavily on how good the method is that is used for generating random points in the circle.

Without loss of generality, we can assume that the circle is $x^2 + y^2 \leq 1$. Letting ξ_1, ξ_2 be a pair of independent random variables distributed uniformly on $[0, 1]$, we let

Table 14.1 The data for Example 14.2.

Fire i	Day x_i	$F_n(x_i)$ $= i/30$	$F(x_i)$ $= x_i/365$	Fire i	Day x_i	$F_n(x_i)$ $= i/30$	$F(x_i)$ $= x_i/365$
1	5	0.033	0.014	16	224	0.533	0.614
2	18	0.067	0.049	17	227	0.567	0.622
3	34	0.100	0.093	18	244	0.600	0.668
4	35	0.133	0.096	19	251	0.633	0.688
5	52	0.167	0.142	20	264	0.667	0.723
6	57	0.200	0.156	21	306	0.700	0.838
7	64	0.233	0.175	22	310	0.733	0.849
8	69	0.267	0.189	23	311	0.767	0.852
9	72	0.300	0.197	24	323	0.800	0.885
10	96	0.333	0.263	25	333	0.833	0.912
11	136	0.367	0.373	26	337	0.867	0.923
12	145	0.400	0.397	27	343	0.900	0.940
13	170	0.433	0.466	28	346	0.933	0.948
14	191	0.467	0.523	29	351	0.967	0.962
15	202	0.500	0.553	30	363	1.000	0.995

$x = 2\xi_1 - 1, y = 2\xi_2 - 1$ and then check whether $x^2 + y^2 \leq 1$. If yes, the pair (x, y) is accepted. If not, we sample a new pair (ξ_1, ξ_2) , and proceed in this way until we obtain an accepted pair (x, y) . One can continue this process to generate as many accepted pairs as needed. The distribution of (x, y) is easily seen to be uniform on the circle with radius 1 centered at the origin, as long as ξ_1, ξ_2 are iid uniform on $[0, 1]$.

To “save computer time” it was proposed that when $x^2 + y^2 > 1$, rather than rejecting both x and y , the value of x is kept, and the computer generates a new ξ_2 , transforms it to a new $y = 2\xi_2 - 1$, and tests if $x^2 + y^2 \leq 1$ (i.e., once x is sampled, it is retained, and only y is added to it).

For an estimation of the savings in the computer time, and for the distribution of the resulting points (x, y) , see Problem 14.3.3. To test the uniformity of the distribution, $n = 100$ points are generated on the circle according to a “time-saving” scheme. Two tests are performed, both reducing the problem to a one-dimensional Kolmogorov–Smirnov test. First, the angles are measured from an arbitrary direction; the angles between this direction and the line connecting point (x, y) with the origin are uniformly distributed on $[0, 2\pi]$. Next, the positive x -axis is the direction chosen, and, counting counterclockwise, the values $\varphi_i = \arctan(x_i/y_i) + (\pi/2)[1 - \text{sign}(y_i)]$ take on a $U[0, 2\pi]$ distribution. The ordered values φ_i are shown in the left part of Table 14.2.

The empirical cdf of these values increases by 0.01 at each $\varphi_{i:100}$, while $F(t) = t/2\pi$. For our data, $\sqrt{n}D_n = \sqrt{n} \sup |F_n(t) - F(t)|$ equals 1.136, which corresponds to a p -value of about 0.15 (see Table B2).

On the other hand, if the points have distribution uniform on the circle with radius 1, the distance R of a random point from the center has a cdf $F_R(t) = P\{R \leq t\} = \pi t^2 / \pi = t^2$ for $0 \leq t \leq 1$. Thus the theoretical cdf of distance R is a parabola t^2 . The right side of Table 14.2 gives the ordered distances. We can then compute the values of the Kolmogorov–Smirnov statistic $\sqrt{n}D_n = \sqrt{n} \sup |F_n(t) - t^2|$, which equals 1.737; the corresponding p -value (see Table B2) is less than 0.005.

Table 14.2

$\varphi_{i:100}$			$R_{i:100}$		
0.0537	1.7784	3.7217	0.0432	0.6757	0.8811
0.0570	1.8854	3.8933	0.1238	0.6969	0.8827
0.0571	1.8967	3.9110	0.1411	0.7045	0.8834
0.1170	1.9798	3.9215	0.1872	0.7145	0.8846
0.1242	2.0339	4.0757	0.2365	0.7168	0.9027
0.1525	2.0721	4.0778	0.2774	0.7268	0.9104
0.3353	2.1705	4.1081	0.3865	0.7312	0.9134
0.4496	2.1833	4.1494	0.4014	0.7326	0.9169
0.4866	2.2170	4.1922	0.4036	0.7365	0.9200
0.5360	2.3099	4.3407	0.4125	0.7462	0.9223
0.5655	2.3482	4.5311	0.4262	0.7547	0.9254
0.5957	2.4588	4.5437	0.4457	0.7557	0.9278
0.6237	2.4618	4.5571	0.4905	0.7833	0.9333
0.6343	2.5279	4.7786	0.4966	0.7836	0.9349
0.7253	2.5874	4.8246	0.5178	0.7924	0.9421
0.8785	2.5840	5.0184	0.5298	0.7967	0.9443
0.8945	2.8752	5.2162	0.5351	0.8175	0.9460
0.9055	2.9165	5.2770	0.5371	0.8268	0.9468
0.9839	2.9489	5.3605	0.5701	0.8290	0.9494
1.0182	2.9534	5.3661	0.5729	0.8311	0.9496
1.1167	2.9606	5.4251	0.5734	0.8337	0.9503
1.1671	2.9669	5.6105	0.5788	0.8412	0.9536
1.2113	2.9774	5.7050	0.5865	0.8453	0.9566
1.3437	3.0322	5.8654	0.5888	0.8472	0.9591
1.4048	3.0663	5.8959	0.6185	0.8476	0.9616
1.4319	3.1596	6.0020	0.6232	0.8485	0.9621
1.4435	3.2035	6.0355	0.6251	0.8491	0.9687
1.4962	3.3165	6.1418	0.6261	0.8521	0.9709
1.5223	3.3226	6.1496	0.6279	0.8632	0.9795
1.5231	3.4023	6.1781	0.6355	0.8656	0.9815
1.5716	3.4204	6.2117	0.6509	0.8669	0.9880
1.5774	3.4407	6.2705	0.6556	0.8692	0.9988
1.6020	3.4981		0.6578	0.8713	
1.7543	3.5595		0.6717	0.8724	

To find the correct p -value of the observed result, we can perform two tests, one giving nonsignificant result corresponding to a p -value 0.15 and the other giving a highly significant result with a p -value of 0.005.

To see what is involved here, suppose that instead of the p -value we just carry out two tests on significance levels α_1 and α_2 , respectively. We decide to *reject* H_0 if either test 1 or test 2 rejects it. Consequently the probability of type 1 error equals

$$\begin{aligned}\alpha &= P\{\text{test 1 rejects or test 2 rejects}|H_0\} \\ &= P\{\text{test 1 rejects}|H_0\} + P\{\text{test 2 rejects}|H_0\} - P\{\text{both tests reject}|H_0\} \\ &= \alpha_1 + \alpha_2 - P\{\text{both tests reject}|H_0\}.\end{aligned}$$

If test 1 and test 2 are performed on two independent samples, then the attained significance level is

$$\alpha = \alpha_1 + \alpha_2 - \alpha_1\alpha_2. \quad (14.8)$$

In the present case, the testing was performed on the same data set. Normally the determination of a significance level in such case is difficult because it requires knowledge of the joint distribution of the statistics used in both tests 1 and 2.

In the case under consideration, however, the situation is simple. Since φ and R are independent (under null hypothesis), we can apply (14.8), which gives the result 0.154.

Two-Sample Kolmogorov–Smirnov Test

In a two-sample problem we need to determine whether or not two samples come from populations with the same distribution. We have already encountered its special cases (e.g., in considering the Student's t test for equality of two normal distributions with the same variance). We now consider this problem in its generality, under the only assumption that the samples are drawn from continuous distributions. Thus we let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two independent random samples from distributions with continuous cdf's F and G , respectively. The objective is to test the hypothesis

$$H_0 : F = G \quad \text{against} \quad H_1 : F \neq G.$$

Unlike in the one-sample case of Section 14.3, the null hypothesis is now composite. The following theorem, also due to Kolmogorov and Smirnov, is presented without proof.

Theorem 14.3.2 *Let $\mathbf{X}_m = (X_1, \dots, X_m)$ and $\mathbf{Y}_n = (Y_1, \dots, Y_n)$ be random samples from distributions with continuous cdf's F and G , respectively, and let $F_m(t)$ and $G_n(t)$ be the respective empirical cdf's. Furthermore let*

$$D_{n,m} = \sup_t |F_m(t) - G_n(t)|.$$

Then

$$\lim_{m,n \rightarrow \infty} P \left\{ \sqrt{\frac{mn}{m+n}} D_{m,n} \leq t \right\} = Q(t),$$

with $Q(t)$ given by (14.7).

If H_0 is true, then the statistic $\sqrt{\frac{mn}{m+n}} D_{m,n}$ has a limiting distribution given by the cdf $Q(t)$, regardless of the particular cdf that governs the sampling of X_i 's and Y_j 's. Suppose

now that H_0 is false. As $m, n \rightarrow \infty$, we have $F_m(t) \rightarrow F(t)$ and $G_n(t) \rightarrow G(t)$ almost surely and uniformly in t , by the Glivenko–Cantelli theorem. Then

$$\begin{aligned} D_{m,n} &= \sup_t |F_m(t) - G_n(t)| \\ &= \sup_t |F_m(t) - F(t) + F(t) - G(t) + G(t) - G_n(t)| \\ &\leq \sup_t |F_m(t) - F(t)| + \sup_t |F(t) - G(t)| + \sup_t |G_n(t) - G(t)|, \end{aligned}$$

and we can write

$$\begin{aligned} \sqrt{\frac{mn}{m+n}} D_{m,n} &\leq \sqrt{\frac{m}{1+\frac{m}{n}}} \sup_t |F_m(t) - F(t)| + \sqrt{\frac{mn}{m+n}} \sup_t |F(t) - G(t)| \\ &\quad + \sqrt{\frac{n}{1+\frac{n}{m}}} \sup_t |G_n(t) - G(t)| \\ &\leq \sqrt{m} \sup_t |F_m(t) - F(t)| + \sqrt{\frac{mn}{m+n}} \sup_t |F(t) - G(t)| \\ &\quad + \sqrt{n} \sup_t |G_n(t) - G(t)|. \end{aligned}$$

The two extreme terms in the last expression have limiting distributions, while the middle term tends to infinity for any single hypothesis contained in the alternative. Thus, again, the test rejects the null hypothesis when the value of the statistic $\sqrt{mn/(m+n)} D_{m,n}$ is large enough.

■ EXAMPLE 14.4

Is a Poisson process observed at every other event also a Poisson process?

SOLUTION. We know, of course, that the answer is negative: If we observe every other event in a Poisson process, then the inter-event times are sums of two exponential random variables and hence are not exponential. Let us verify this fact empirically. Table 14.3 gives $m = 12$ interarrival times in a Poisson process with mean 1 (where every event is observed) and $n = 16$ interarrival times for every other event in another Poisson process with mean $\frac{1}{2}$ (so the mean interarrival times are the same).

Column T_i gives the observed interarrival times in both samples, jointly ordered, while the next two columns give the values of $F_m(t)$ and $G_n(t)$ at the observed points— $F_m(t)$ increases at points from the first sample while $G_n(t)$ increases at points from the second sample. It may be seen therefore that the six shortest interarrival times are all in a Poisson process observed at every event. Here the value $D_{m,n}$ is 0.5 and the statistic $\sqrt{mn/(m+n)} D_{m,n}$ equals 1.309, corresponding to a p -value of about 0.065.

PROBLEMS

14.3.1 Let the observed values $x_1, \dots, x_n$ be such that $1/3 \leq x_i \leq 2/3$ for all i . What can be said about n , if the null hypothesis that the underlying distribution is $U[0, 1]$ cannot be rejected by the Kolmogorov–Smirnov test at $\alpha = 0.05$ level?

Table 14.3

T_i	$F_m(\cdot)$	$G_n(\cdot)$	T_i	$F_m(\cdot)$	$G_n(\cdot)$
0.049	0.083	0.000	0.942	0.750	0.375
0.198	0.166	0.000	0.969	0.750	0.437
0.237	0.250	0.000	1.033	0.833	0.437
0.259	0.333	0.000	1.094	0.833	0.500
0.310	0.416	0.000	1.375	0.833	0.562
0.352	0.500	0.000	1.392	0.833	0.625
0.381	0.500	0.062	1.555	0.833	0.687
0.546	0.500	0.125	1.625	0.833	0.750
0.547	0.500	0.187	1.697	0.833	0.812
0.569	0.583	0.187	2.019	0.916	0.812
0.801	0.583	0.250	2.065	1.000	0.812
0.803	0.666	0.250	2.114	1.000	0.875
0.878	0.750	0.250	2.244	1.000	0.937
0.895	0.750	0.312	2.534	1.000	1.000

- 14.3.2** Suppose that the data are as in Example 14.2, except that there were only 25 fires, none of them in November. Test that the fires occur according to the Poisson process.
- 14.3.3** Find the joint density of (X, Y) resulting from the “time-saving scheme” of Example 14.3. Find also the expected number of random variables ξ_i necessary to sample in order to obtain one pair (X, Y) under both schemes.
- 14.3.4** What is the minimal possible value of the statistic $D_{m,n}$ if k values of X_i precede the third in the magnitude value Y_j ? If $m = 100$, $k = 30$, and $n = 200$, is there enough evidence to reject (at the level $\alpha = 0.05$) the null hypothesis that the distributions of X ’s and Y ’s are the same?
- 14.3.5** Suppose that out of 30 fires in Example 14.2 those on January 5 and 18, February 3 and 21, March 10, April 6, May 25, June 19, December 3 were caused by arson, and in the remaining cases arson was excluded. Use the Kolmogorov–Smirnov statistic to test the hypothesis that the occurrences of “arson” and “nonarson” fires within a year follow the same distribution.
- 14.3.6** Suppose that one sample contains $2m$ data points while the other contains $2m + k$ data points. The first $2m$ and the last $2m$ data points in the joint sample alternate between samples. Thus the ordered data has the form

$$\underbrace{YX \cdots YX}_{2m} \underbrace{YY \cdots Y}_{k} \underbrace{XY \cdots XY}_{2m}. \quad (14.9)$$

(i) For given m find k such that the Kolmogorov–Smirnov test will reject the hypothesis that both samples are drawn from the same population ($\alpha = 0.05$). (ii) Solve this problem if the string of k consecutive Y ’s occurs at the beginning of the joint ordering.

14.3.7 Assume that each of two samples contains $2m + k$ elements with the following ordering:

$$\underbrace{YX \cdots YX}_{2m} \underbrace{YY \cdots YY}_k \underbrace{XY \cdots XY}_{2m} \underbrace{XX \cdots XX}_k \quad (14.10)$$

For given m find k such that the Kolmogorov–Smirnov test will reject the hypothesis that both samples are drawn from the same population ($\alpha = 0.05$).

14.4 ONE-SAMPLE RANK TESTS

We begin this section with the Wilcoxon signed rank test. This test, used for testing hypotheses about a location parameter in symmetric distributions (median), is an excellent example of the simplicity and versatility of nonparametric methods.

Assume that we have a random sample $X_1, \dots, X_n$ from a continuous, symmetric distribution with a cdf F and a density f . Then there exists θ such that for any x ,

$$F(\theta - x) = 1 - F(\theta + x).$$

Equivalently we can say that $f(\theta + x) = f(\theta - x)$ for any x .

The null hypothesis $H_0 : \theta = \theta_0$ will be tested against either the one-sided alternative $H_1 : \theta > \theta_0$ or the two-sided alternative $H_1 : \theta \neq \theta_0$. We can use this test also for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. The case of null and alternative hypotheses involving opposite inequalities can be obtained by an obvious change of signs. To define the test statistic, consider the absolute differences $V_1 = |X_1 - \theta_0|, V_2 = |X_2 - \theta_0|, \dots, V_n = |X_n - \theta_0|$. Since the underlying distribution is continuous, we can assume that all V_i 's are distinct and that none equals 0.

Let us arrange the V_i 's in increasing order and assign ranks $R_1, R_2, \dots, R_n$ to them, with rank 1 assigned to the smallest V_i . Furthermore let

$$\eta_i = \begin{cases} +1 & \text{if } X_i > \theta_0 \\ -1 & \text{if } X_i < \theta_0. \end{cases}$$

The *Wilcoxon signed rank statistic* is defined as

$$S_n = \sum_{i=1}^n \eta_i R_i.$$

We construct a test for any of the hypotheses mentioned above, we need to:

1. Find the distribution (or at least, limiting distribution as n becomes large) of the statistic S_n under the null hypothesis.
2. Study the effect of values of θ in the alternative hypothesis on the values of S_n .

Suppose therefore that the true value of θ is θ_0 . In this case the signs η_i are equally likely to be positive or negative:

$$P\{\eta_i = 1\} = P\{\eta_i = -1\} = \frac{1}{2}.$$

Moreover, the random variables $\eta_1, \dots, \eta_n$ are independent (since each is determined by a different X_i), and also η_i is independent of R_i , by symmetry of the distribution of X_i about θ_0 . The values $R_1, \dots, R_n$ form a permutation of numbers $1, \dots, n$, so we write

$$S_n = \sum_{i=1}^n i \eta_i.$$

Since $E(\eta_i) = 0$ and $\text{Var}(\eta_i) = E(\eta_i^2) = 1$, we also have $E(S_n) = 0$ and

$$\text{Var}(S_n) = \sum_{i=1}^n i^2 \text{Var}(\eta_i) = \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

It is possible, though tedious, to determine the distribution of S_n for small values of n ; the exact distribution of S_n can be found in almost any sufficiently large collection of statistical tables. For large n one can prove that $S_n/\sqrt{\text{Var}(S_n)}$ has a limiting standard normal distribution.

Indeed, S_n is the sum of n independent random variables $\eta_1 + 2\eta_2 + \cdots + n\eta_n$, where $i\eta_i = \pm i$ with probability $1/2$ each. To apply the Lindeberg–Feller theorem (9.6.5), we have $\text{Var}(i\eta_i) = E[(i\eta_i)^2] = i^2$, so $s_n^2 = \text{Var}(S_n) = \sum_{i=1}^n i^2 \sim n^3$. Since $E(i\eta_i) = 0$ for $i = 1, 2, \dots$, we have to show that

$$\frac{1}{s_n^2} \sum_{i=1}^n \int_{|x| \geq \epsilon s_n} x^2 dF_i(x) \rightarrow 0$$

for every $\epsilon > 0$, where F_i is the cdf of $i\eta_i$. Since $s_n \sim n^{3/2}$, we have $\epsilon s_n > n$ for n large enough, and each integral equal to 0; since $i\eta_i$ is either i or $-i$, we have $|i\eta_i| \leq n$. This shows that Lindeberg–Feller condition is satisfied, and the proof is complete.

To determine now whether to use the right tail, left tail, or both tails of the distribution of S_n (limiting or exact), observe that if the true value of θ exceeds θ_0 , then

$$P\{\eta_i = +1\} = P\{X_i > \theta_0\} > P\{X_i > \theta\} = \frac{1}{2}.$$

Consequently, the positive signs are more likely than negative signs, and this will tend to increase the value of S_n . Thus, in tests of $H_0 : \theta = \theta_0$ or $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$, large values of S_n provide evidence for the alternative, and the right tail should be used as the critical region.

■ EXAMPLE 14.5

Assume that f is a density symmetric about 0; that is, it satisfies the condition $f(x) = f(-x)$ for all x . Let $X_1, \dots, X_n$ be a random sample from the distribution with density f , and suppose that we observe the values $Y_i = \theta + X_i, i = 1, \dots, n$. Thus θ is the median of Y_i (and also its mean, if $E|X_i| < \infty$). Let the observed values of $n = 20$ observations of Y_i be 315, 493, 366, 291, 501, 503, 388, 526, 308, 410, 418, 540, 285, 360, 426, 475, 336, 455, 301, 359. We want to test the hypothesis that the median θ satisfies the inequality $\theta \leq 350$, against the alternative $\theta > 350$. The consecutive values $|Y_i - 350|$, the signs η_i of the differences $Y_i - 350$, and ranks R_i are listed in Table 14.4.

The value of statistic S_{20} equals 124. The asymptotic variance of S_{20} is $20 \times 21 \times 41/6 = 2,870$, and therefore the observed value of statistic $S_{20}/\sqrt{\text{Var}(S_{20})}$ is 2.31. The corresponding p -value is 0.0129, so we have strong evidence in the data that the median of the population exceeds 350.

The Wilcoxon one-sample test can also be used to compare two related populations. This setup was already introduced in Chapter 12, in the test we developed for the mean difference in paired data.

■ EXAMPLE 14.6 Paired Data

When comparing the effect of two treatments A and B , part of the variation in the data is caused by other factors—for example in medical experiments this could be patient's

Table 14.4

i	$ Y_i - 350 $	η_i	Rank	i	$ Y_i - 350 $	η_i	Rank
1	35	-1	5	11	68	+1	12
2	143	+1	16	12	190	+1	20
3	16	+1	4	13	65	-1	11
4	59	-1	9	4	10	+1	2
5	151	+1	17	15	76	+1	13
6	153	+1	18	16	525	+1	15
7	38	+1	6	17	14	-1	3
8	176	+1	19	18	105	+1	14
9	42	-1	7	19	49	-1	8
10	60	+1	10	20	9	+1	1

age, gender, health status, and so on. To eliminate, or at least significantly reduce that variability, the experimenter should apply treatments A and B to the same, or at least very similar (“almost the same”) subjects. Therefore our data will concern pairs, either formed naturally (e.g., twins) or carefully matched as closely as possible (age, gender, etc.). It is important to realize that the pairing process is not random. In fact, the strength of the final conclusion depends largely on how well the members of each pair are matched by as many factors as possible.

Next within each pair the experimenter allocates one member to treatment A , and the other to treatment B , this allocation being random. Let X_i and Y_i denote the results of treatments A and B , respectively, for members of the i th pair of subjects, and let $Z_i = X_i - Y_i$. If the treatments do not differ, then Z_i is as likely to be positive as negative (i.e., its median is zero). It will have a symmetric distribution because members of the pair are matched and treatments are allocated at random to members of the pair. Thus the hypothesis “treatments do not differ” and “treatment A is superior” are now expressed as $H_0 : \theta = 0$ and $H_1 : \theta > 0$. The Wilcoxon signed rank procedure can be used to test these hypotheses.

We will present here one more procedure, called the runs test. In the case of a single sample, it can be used to test the hypothesis that the sample elements were selected randomly. In the case of two samples, it can be used to test the equality of two underlying distributions. For example, suppose that in a sample of size $n = 10$, we observe that the first five values are all negative while the last five values are all positive. Is this an indication that the process of taking the sample was not random?

The general idea of the runs test is to partition the observations $X_1, \dots, X_n$ into two classes, say A and $B = A^c$, in such a way that the partition is induced by the values of X_i only, not by their order (A may be the set of observations that are positive, observations that exceed the hypothetical median, etc.). Formally this means that if the vector $(X_1, \dots, X_n)$ leads to a choice of observations with indices $i_1, \dots, i_k$ to form the set A and $\pi(1), \dots, \pi(n)$ is a permutation of indices $(1, \dots, n)$, then the choice of set A from the permuted vector of observations $(X_{\pi(1)}, \dots, X_{\pi(n)})$ is the set with indices $\pi(i_1), \dots, \pi(i_k)$. The randomness of the sample, combined with the fact that with probability 1 there are no ties among the sample values, imply that each set of indices of appropriate size is equally likely to be the set A of the partition.

In the special case we could have two samples, $X_1, \dots, X_m$ from a distribution F and $Y_1, \dots, Y_n$ from a distribution G , and we would test the null hypothesis that $F \equiv G$ (i.e., samples come from the same distribution). We would then consider the joint sample of X 's and Y 's and the partition would correspond to the two constituent samples. In this case, if the null hypothesis is true, each arrangement of X 's and Y 's would be equally likely.

The test is now based on the intuitive idea that if the sampling is really random (or, in the two-sample case, if samples come from the same distribution), then the partition into sets A and $B = A^c$ is "random." One of the possible measures of deviations from randomness is to observe the number of runs. To fix the idea, imagine that we have $m = 5$ symbols A and $n = 7$ symbols B arranged in some order, such as

$$\underline{BBABAAABBBAB}.$$

A run is a string of elements of one kind, bordered either by elements of the other kind or by the end of the string. For instance, the string above has four runs of elements B , as underlined, and three runs of elements A , not underlined. The test is based on the intuitive expectation that a too small number of runs, such as

$$AAAAABBBBBBB,$$

or a too large number of runs, such as

$$BABABABABAB,$$

indicates lack of randomness.

To develop the test, we need the distribution of the number of runs under the null hypothesis. Let R be the total number of runs. In the sequel we will assume that $m \leq n$. Clearly, the smallest value of R is 2, while the largest possible value of R is obtained for alternating runs of A 's of length 1. If $m < n$, then the maximal value of R is $2m + 1$, while if $m = n$, then it is $2m$.

It remains to determine the probabilities $P\{R = r\}$ for all possible values of r . Consider first the case where r is even ($r = 2k$). The sequence must then contain k runs of each kind that alternate, starting either with a run of A 's or with a run of B 's. Let us imagine m elements A arranged in a string. Dividing the string into k runs means choosing $k - 1$ out of $m - 1$ places separating consecutive A 's. This can be done in $\binom{m-1}{k-1}$ ways. In a similar way the string of B 's of length n can be divided into k runs in $\binom{n-1}{k-1}$ distinct ways.

A joint string with $2k$ runs is now formed by dividing A 's and B 's into k runs each, as described above, and joining them by taking alternating runs. For instance, suppose $k = 3$, $m = 5$, and $n = 7$. The string of five A 's can be partitioned into three runs in $\binom{4}{2} = 6$ ways:

$$AAA|A|A \quad AA|AA|A \quad AA|A|AA \quad A|A|AAA \quad A|AAA|A \quad A|A|AAA.$$

In a similar way, a string of seven B 's can be divided into 3 strings in $\binom{6}{2} = 15$ ways. Taking one such partition for A 's and one for B 's, we obtain two arrangements giving $r = 2 \times 3 = 6$ runs. For example, taking the partition $AAA|A|A$ and $B|BBBB|BB$ we obtain two arrangements:

$$AAABABBBBABBB \quad \text{and} \quad BAAABBBBABBA.$$

Consequently

$$P(R = 2k) = \frac{2 \binom{m-1}{k-1} \binom{n-1}{k-1}}{\binom{m+n}{m}}, \quad (14.11)$$

where the denominator gives the total number of arrangements of m objects A and n objects B . For $r = 2k + 1$ we must have either k runs of A 's and $k + 1$ runs of B 's, or vice versa. Reasoning analogous to that used in obtaining (14.11) leads to

$$P(R = 2k + 1) = \frac{\binom{m-1}{k-1} \binom{n-1}{k} + \binom{m-1}{k} \binom{n-1}{k-1}}{\binom{m+n}{m}}. \quad (14.12)$$

Formulas (14.11) and (14.12) therefore give the distribution of the number of runs R .

■ EXAMPLE 14.7

A machine produces items whose nominal diameter is c . Because of inherent variability the diameters of the items produced are random, sometimes above c and sometimes below it. The machine was designed in such a way that the diameter of the each item produced has no effect on the diameter of the next one. The diameters of 12 consecutively produced items were recorded and classified as “above c ” (A) or “below c ” (B). The resulting sequence was $BAAAAABBBBBB$, so $m = 5$, $n = 7$, and $R = 3$. The small number of runs led to the suspicion that there may be some systematic low-frequency oscillation in operation of the machine, which tends to produce long runs of items with dimensions above c , followed by long runs of items with dimensions below c .

To test the hypothesis on “randomness” of dimensions of items against the alternative of “low-frequency oscillation,” we must choose the left tail of the distribution of R (low number of runs) as the critical region, or equivalently, determine the p -value of the observed result, that is, $P(R \leq 3)$. In this case the calculations are straightforward: using formulas (14.11) and (14.12), we obtain, for $m = 5$, $n = 7$, and $k = 1$,

$$\begin{aligned} P(R \leq 3) &= P(R = 2) + P(R = 3) \\ &= \frac{1}{\binom{12}{5}} \left\{ 2 \binom{4}{0} \binom{6}{0} + \left[\binom{4}{0} \binom{6}{1} + \binom{4}{1} \binom{6}{0} \right] \right\} \\ &= \frac{2 + 6 + 4}{\binom{12}{5}} = \frac{12}{792} = 0.0152. \end{aligned}$$

Thus observing only three runs in this situation is a strong indication of the low-frequency oscillation effect.

Calculations such as in the example above are cumbersome for large m and n . Fortunately, we can use the normal approximation showing first that

$$E(R) = \frac{2mn}{m+n} + 1 \quad \text{and} \quad \text{Var}(R) = \frac{2mn(2mn - m - n)}{(m+n)^2(m+n-1)}.$$

We also have the following theorem:

Theorem 14.4.1 *If $m \rightarrow \infty$, $n \rightarrow \infty$ in such a way that $m/n \rightarrow \eta$ with $0 < \eta < \infty$, then*

$$\frac{R - 2m/(1 + \eta)}{\sqrt{4\eta m/((1 + \eta)^3)}}$$

converges in distribution to $N(0, 1)$.

Thus, replacing η by m/n , we can expect that the random variable

$$\frac{R - 2mn/(m+n)}{2mn} \sqrt{(m+n)^3} \quad (14.13)$$

has approximate standard normal distribution, provided that m and n are large.

We can show that the approximation is very good for $m, n \geq 20$, and quite acceptable when $m, n \geq 10$.

■ EXAMPLE 14.8

Recall Example 14.7. Using (14.13) we have an approximation

$$\begin{aligned} P\{R \leq 3\} &= P\left\{\frac{R - 2 \times 5 \times 7/(5+7)}{2 \times 5 \times 7} \sqrt{(5+7)^3}\right\} \\ &\leq \frac{3 - 2 \times 5 \times 7/(5+7)}{2 \times 5 \times 7} \sqrt{(5+7)^3}\bigg\} \\ &\approx P\left\{Z \leq \frac{3 - 5.83}{70} \times 41.57\right\} = P\{Z \leq -1.68\} = 0.0465. \end{aligned}$$

As compared with the exact p -value 0.0152, the approximation is not good.

This shows that for small sample sizes one should try to determine the exact p -value (by direct evaluation, use of special tables, or an appropriate statistical package).

PROBLEMS

- 14.4.1** Prove the asymptotic normality of the Wilcoxon signed rank statistic S_n using the Liapunov theorem.
- 14.4.2** Out of 15 data points one is between 0 and 1, two are between -2 and -1 , three are between 2 and 3, four are between -4 and -3 , and five are between 4 and 5. Use the Wilcoxon signed rank statistic to test the hypothesis that the median is: (i) 0. (ii) 1.
- 14.4.3** Twelve pairs of subjects, matched within each pair with respect to age, gender, health status, and initial weight, were put on two types of diets. The data on pounds lost after 5 weeks are as follows:

Pair	1	2	3	4	5	6	7	8	9	10	11	12
Diet A	15	33	21	17	14	25	25	31	18	5	46	11
Diet B	18	17	10	10	32	11	8	26	-3	19	5	8

Use the Wilcoxon signed ranked test to test the hypothesis that both diets have the same effect, against the alternative that diet B is more efficient than diet A . Use $\alpha = 0.05$.

- 14.4.4** Some texts define the Wilcoxon signed rank statistic as $\tilde{S}_n = \sum \tilde{\eta}_i R_i$, where $\tilde{\eta}_i = 1$ if $X_i > \theta_0$ and 0 otherwise. Determine the mean and variance of $\tilde{S}_n$, and show that tests based on S_n and on $\tilde{S}_n$ are equivalent in the following sense: under a null hypothesis, $S_n = \tilde{S}_n - \tilde{S}_n^*$, where $\tilde{S}_n$ and $\tilde{S}_n^*$ have the same distribution.

- 14.4.5** Solve Problems 14.3.6 and 14.3.7 using a runs test. Compare the results obtained by different methods (use the same $\alpha = 0.05$). Explain the differences, if they exist, for $m = 10$ and $k = 5$.
- 14.4.6** A machine is set up to produce items, each with a diameter above 1 inch. The diameters of 15 consecutive items produced are 1.11, 1.15, 0.98, 1.11, 1.08, 1.06, 0.97, 0.97, 1.05, 1.02, 0.98, 0.99, 0.96, 1.03, 1.01. Use a runs test, taking 1 inch as a threshold to test the hypothesis that the measurements represent random deviations from the required standard.

14.5 TWO-SAMPLE RANK TESTS

Let $\mathbf{X} = (X_1, \dots, X_m)$ and $\mathbf{Y} = (Y_1, \dots, Y_n)$ be random samples from two continuous distributions with cdf's F and G , respectively. We want to test the null hypothesis

$$H_0 : F \equiv G$$

against some alternatives, whose form will be discussed later.

We present one of the most important of the rank tests, the Wilcoxon–Mann–Whitney two-sample test. The underlying idea of the test is as follows:

First, we combine both samples and then arrange all observations in an increasing order:

$$U_1 \leq U_2 \leq \dots \leq U_{m+n},$$

where each U_i belongs to one of the samples. Because of the assumed continuity of F and G , we can disregard the possibility of ties in the joint sample, so we assume that all inequalities among U_i 's are strict.

Next we assign ranks from 1 to $m + n$ to consecutive elements U_i . If the null hypothesis is true, then the m ranks of elements of the first sample and the n ranks of elements of the second sample are mixed randomly, in the sense that each of the $\binom{m+n}{m}$ allocations of the m ranks of elements in the first sample has the same probability $1/\binom{m+n}{m}$.

Wilcoxon suggested to use the statistic W_X , defined as the sum of ranks of elements of the sample $(X_1, \dots, X_m)$, in the joint ordering of both samples. Formally, we can write

$$W_X = \sum_{i=1}^{m+n} i I_X(U_i),$$

where $I_X(U_i) = 1$ if U_i comes from sample $\mathbf{X}$ and is 0 otherwise.

Equivalently we can use statistic W_Y , being the sum of ranks of elements of the second sample in the joint ordering. Statistics W_X and W_Y carry the same information. Indeed we have

$$\begin{aligned} W_X + W_Y &= \sum_{i=1}^{m+n} i I_X(U_i) + \sum_{i=1}^{m+n} i I_Y(U_i) = \sum_{i=1}^{m+n} i (I_X(U_i) + I_Y(U_i)) \\ &= \sum_{i=1}^{m+n} i = \frac{(m+n)(m+n-1)}{2}, \end{aligned}$$

so

$$W_Y = \frac{(m+n)(m+n-1)}{2} - W_X.$$

Let us begin with finding the expectation and variance of W_X . We have

$$E(W_X) = E\left(\sum_{i=1}^{m+n} iI_X(U_i)\right) = \sum_{i=1}^{m+n} iE[I_X(U_i)] = \sum_{i=1}^{m+n} iP\{I_X(U_i) = 1\}.$$

Since all allocations of the m ranks of elements of the first sample among $m+n$ elements of both samples are equally likely, the probability that the i th ranking element comes from the first sample is $P(I_X(U_i) = 1) = m/(m+n)$. We have therefore

$$E(W_X) = \sum_{i=1}^{m+n} i \frac{m}{m+n} = \frac{m(m+n+1)}{2}.$$

The calculations of variance is somewhat more tedious (we leave it as exercise). The result is

$$\text{Var}(W_X) = \frac{mn(m+n+1)}{12}. \quad (14.14)$$

One could also show (we omit the proof) that as $m \rightarrow \infty, n \rightarrow \infty$, the random variable

$$Z_{m,n} = \frac{W_X - E(W_X)}{\sqrt{\text{Var}W_X}} = \frac{W_X - m(m+n+1)/2}{\sqrt{mn(m+n+1)/12}}$$

converges in distribution to the standard normal random variable.

To design a testing procedure, it is now necessary to specify the alternative hypothesis. In other words, we have to determine the class of hypotheses such that if (F, G) belongs to this class, then the values of $Z_{m,n}$ will tend to be relatively large (or small, or large in absolute value).

One such class of alternative hypotheses is obtained by taking

$$H_1 : G(t) = F(t - \theta) \quad (14.15)$$

for all t and some θ . To grasp the meaning of this hypothesis and its consequence for W_X , let us consider the case $\theta > 0$. Here $G(t) = F(t - \theta) \leq F(t)$, which means that $P\{Y \leq t\} \leq P\{X \leq t\}$; hence the values of X tend to be smaller than values of Y (since whatever the value t , the random variable X is more likely to be below t than the random variable Y). Consequently the observations $X_1, \dots, X_m$ will tend to be located closer to the left end and so have smaller ranks. Thus small values of W_X support the alternative hypothesis $H_1 : \theta > 0$. The case $\theta < 0$ is analogous, whereas for the alternative $H_1 : G(t) = F(t - \theta), \theta \neq 0$, one should take the two-sided test.

Actually the class of alternatives against which the Wilcoxon statistic W_X can be used is larger. Recall that the random variable X is *stochastically larger* than Y if $P\{X \leq t\} \leq P\{Y \leq t\}$ for all t . In the present notation, X is stochastically larger than Y if $F(t) \leq G(t)$ for all t . Thus stochastic dominance of Y by X (or X by Y) is an alternative that will tend to inflate (or decrease) the statistic W_X .

About the same time as Wilcoxon introduced his statistic W_X , Mann and Whitney introduced another statistic, pertaining to the same two-sample problem:

$$R = \text{number of pairs } (X_i, Y_j) \text{ with } X_i > Y_j.$$

The statistic W_X can be written as

$$W_X = \sum_{j=1}^m R_j,$$

where R_j is the rank of $X_{j:m}$. This means that R_j equals the number of elements in the combined sample that do not exceed $X_{j:m}$.

By definition, there are j elements in the sample $X_1, \dots, X_m$ that are less or equal to $X_{j:m}$, so we can write

$$R_j = j + \text{number of } Y_i \text{ with } Y_i < X_{j:m}.$$

Consequently

$$W_X = \sum_{j=1}^m [j + \text{number of } i\text{'s with } Y_i < X_{j:m}] = \frac{m(m+1)}{2} + R.$$

It follows that

$$E(R) = E(W_X) - \frac{m(m+1)}{2} = \frac{m(m+n+1)}{2} - \frac{m(m+1)}{2} = \frac{mn}{2}$$

and

$$\text{Var}(R) = \text{Var}(W_X) = \frac{mn(m+n+1)}{2}.$$

It also follows, from the asymptotic normality of W_X , that as $m \rightarrow \infty, n \rightarrow \infty$, the statistic

$$Z_{m,n} = \frac{R - mn/2}{\sqrt{mn(m+n+1)/2}} \quad (14.16)$$

converges in distribution to $N(0, 1)$. Since the tests based on W_X and on R are equivalent, they became known as Wilcoxon–Mann–Whitney tests.

Observe now that there are mn possible pairs with elements of the pair coming from different samples. Consequently, R/mn is a consistent estimator of the probability $P\{Y < X\}$, so as $m \rightarrow \infty, n \rightarrow \infty$, we have

$$\frac{R}{mn} \xrightarrow{p} P\{Y < X\} = \int_{-\infty}^{+\infty} G(t)f(t)dt = \int_{-\infty}^{+\infty} [1 - F(t)]g(t)dt.$$

Under the null hypothesis $F \equiv G$, we have $P\{Y < X\} = 1/2$. If $P\{Y < X\} = \xi \neq 1/2$, then, using (14.16), we can write

$$\begin{aligned} Z_{m,n} &= \sqrt{\frac{2mn}{m+n+1}} \left(\frac{R}{mn} - \frac{1}{2} \right) \\ &= \sqrt{\frac{2mn}{m+n+1}} \left(\frac{R}{mn} - \xi \right) + \sqrt{\frac{2mn}{m+n+1}} \left(\xi - \frac{1}{2} \right). \end{aligned}$$

The first term converges in distribution to a standard normal random variable, while the second diverges to $+\infty$ or $-\infty$ if only $\xi \neq 1/2$. This shows that in the limit as $m \rightarrow \infty, n \rightarrow \infty$, the Wilcoxon–Mann–Whitney test (based on W_X or R) has asymptotic power 1 for all alternatives (F, G) with $\xi \neq 1/2$.

■ EXAMPLE 14.9

Consider two athletes, A and B ; one of them is to be selected to represent the country in some competition. Assume that both athletes have attained some stable level of proficiency in their discipline. Their results (in competitions, their best daily training results, etc.) may be taken as random samples of some random variables, say $X^{(A)}$ and $X^{(B)}$. Furthermore, assume that the discipline is such that tied results are unlikely. Let

$n = 15$ be the results of A and $n = 20$ the results of B , with none of the results repeating. Suppose that after arranging these results jointly from worst to the best, we obtain the sequence

$A A A B A B B A A A B B B B A B A B B A A B B A B B B A B B B A$

Here the three lowest results are of athlete A , the fourth is of athlete B , and so on. The best result belongs also to A .

We can define the “better” athlete (of a pair) as the one who has better than even chances of defeating the other one. Thus A is better than B if

$$\xi = P(A \text{ beats } B) > \frac{1}{2}.$$

The idea that lies at the foundation of the US system of selecting Olympic representatives is the same as that in the case under consideration: A is better than B because the best result of A is better than the best result of B . Let us therefore test the hypothesis that

$$\xi = P(A \text{ beats } B) = P(X > Y) \leq \frac{1}{2}$$

(A is equal to B , or inferior to B) against the alternative that $\xi > 1/2$. The Wilcoxon statistic (sum of ranks of A) is

$$W_X = 1 + 2 + 3 + 5 + 8 + 9 + 10 + 15 + 17 + 20 + 23 + 24 + 27 + 31 + 35 = 230.$$

For $m = 15, n = 20$ we have $E(W_X) = 270$, $\text{Var}(W_X) = 900$, so the p -value is $P\{Z > (230 - 270)/30\} = P\{Z > -1.33\} = 0.9082$. There is therefore no reason to reject the null hypothesis that A is no better than B , despite the fact that the best result is attained by athlete A .

PROBLEMS

14.5.1 Prove formula (14.14) showing first that for $i \neq j$

$$\text{Var}(I_{A_i}) = \frac{mn}{(m+n)^2} \quad \text{and} \quad \text{Cov}(I_{A_i}, I_{A_j}) = -\frac{mn}{(m+n)^2(m+n-1)}.$$

14.5.2 Two samples of sizes $m = 20$ and $n = 10$, respectively, are selected from two populations. Let $r_1, \dots, r_{10}$, denoting the numbers of observations from the first sample that do not exceed the k th ($k = 1, \dots, 10$) element in the ordered second sample, be 1, 1, 2, 4, 4, 6, 8, 9, 11, d . **(i)** Find the value of Wilcoxon and Mann–Whitney statistic as a function of d . **(ii)** Find the value of Kolmogorov–Smirnov statistic as a function of d . **(iii)** Suggest the appropriate alternative hypothesis, and determine d for which the hypothesis about the same median can be rejected based on the Mann–Whitney or Wilcoxon test. **(iv)** Answer **(iii)** using the Kolmogorov–Smirnov test.

14.5.3 Use the runs test for the data of Problem 14.5.2.

14.5.4 Assume that n is odd and that all m elements of the first sample are below the median of the second sample. Find the range of the test statistics and the rejection region for the appropriate alternative hypothesis using: **(i)** The Wilcoxon test. **(ii)** The Mann–Whitney test. **(iii)** The runs test. **(iv)** The Kolmogorov–Smirnov test for the appropriate alternative hypothesis. ($\alpha = 0.05, n = 51, m = 20$).

14.6 KRUSKAL–WALLIS TEST

Finally, we present the Kruskal–Wallis test, a rank-based counterpart of the one-way ANOVA test. We consider k random samples, of sizes $n_1, n_2, \dots, n_k$, where $k \geq 2$. We let X_{ij} be the j th element ($j = 1, \dots, n_i$) in i th sample, and we assume that $X_{i,1}, \dots, X_{i,n_i}$ are iid random variables with a cdf given by

$$F_i(x) = G(x - \theta_i)$$

for all x . Here, G is assumed to be a cdf of some continuous random variable. We want to test the null hypothesis

$$H_0 : \theta_1 = \theta_2 = \dots = \theta_k \quad \text{against} \quad H_1 : \text{not all } \theta'_i \text{ are equal.}$$

If G is a cdf of a distribution symmetric around 0—satisfying $G(-x) = 1 - G(x)$ —then the median and the mean (if it exists) of F_i is easily seen to be θ_i , and the null hypothesis asserts that the medians (or means) of all populations are the same. The variance need not exist, but the fact that all populations have the same distribution up to a location parameter corresponds to the assumption of homoscedasticity in Chapter 13.

Let us order all $n = n_1 + \dots + n_k$ observations X_{ij} from smallest to largest, and let R_{ij} be the rank of observation X_{ij} in the joint ordering. Furthermore let $R_{i+} = \sum_{j=1}^{n_i} R_{ij}$ be the sum of ranks corresponding to elements in the i th sample (if we have two populations, i.e., $k = 2$, then R_1 and R_2 correspond to Wilcoxon statistics W_X and W_Y from the preceding section).

The test of null hypothesis H_0 is based on Kruskal–Wallis statistic

$$B = \frac{12}{n(n+1)} \sum_{i=1}^r \frac{R_{i+}^2}{n_i} - 3(n+1). \quad (14.17)$$

We have the following theorem:

Theorem 14.6.1 *If H_0 is true, and all sample sizes $n_1, n_2, \dots, n_r$ increase to infinity in such a way that $n_i/n \rightarrow p_i > 0$ for all i , then the distribution of statistic B converges to the chi-square distribution with $r - 1$ degrees of freedom.*

We will not give a full proof, but outline the argument to explain why the large values of the statistic B are compatible more with the alternative than with the null hypothesis.

First, under the null hypothesis,

$$E\left(\frac{R_{i+}}{n_i}\right) = \frac{n+1}{2} \quad \text{and} \quad \text{Var}\left(\frac{R_{i+}}{n_i}\right) = \frac{(n+1)(n-n_i)}{12n_i}. \quad (14.18)$$

The proof is left as an exercise. Now letting

$$V_i = \frac{R_{i+}/n - E(R_{i+}/n)}{\sqrt{\text{Var}(R_{i+}/n)}},$$

we write

$$\begin{aligned}
 \sum_{i=1}^r \left(1 - \frac{n_i}{n}\right) V_i^2 &= \sum_{i=1}^r \left(1 - \frac{n_i}{n}\right) \left[\frac{R_{i+}/n_i - E(R_{i+}/n_i)}{\sqrt{\text{Var}(R_{i+}/n_i)}} \right]^2 \\
 &= \sum_{i=1}^r \frac{n - n_i}{n} \times \frac{(R_{i+}/n_i - (n+1)/2)^2}{(n+1)(n - n_i)/12n_i} \\
 &= \frac{12}{n(n+1)} \sum_{i=1}^r \frac{R_{i+}^2}{n_i} - 3(n+1).
 \end{aligned} \tag{14.19}$$

First, from expression (14.19) we have that any deviation from the null hypothesis will tend to increase B . Thus the critical region is always the right tail of the appropriate chi-square distribution, as in ANOVA tests under normal assumptions.

Second, we have to show that V_i converges in the distribution to a standard normal variable, or equivalently that the distribution of R_{i+} is asymptotically normal. Once this fact is established, the proof of the theorem can be completed by observing that the random variables $R_{1+}, \dots, R_{r+}$ are constrained by the condition $\sum_{i=1}^r R_{i+} = n(n+1)/2$, which reduces the number of degrees of freedom to $r - 1$. It remains to check, for instance, that the asymptotic mean and variance of B agree of those of chi-square distribution with $r - 1$ degrees of freedom.

The proof of asymptotic normality of R_{i+} lies beyond the scope of this book; it relies on one of the central limit theorems for the sum of exchangeable random variables.

PROBLEMS

14.6.1 Prove relations (14.18).

14.6.2 STAT 102 can only be taken by students who passed STAT 101. Among 15 students in STAT 102, five took STAT 101 from instructor X , four took it from instructor Y , and the rest took the course from instructor Z . Ordered according to their performance, the students of the three instructors are

$$ZXZZZYXZYXYXZZXY$$

(this means that the best student was taught by instructor Z , second best by instructor X , etc.). At the significance level $\alpha = 0.05$, test the hypothesis that the performance of students in STAT 102 class does not depend on who taught them STAT 101.